InventOR: Can Large Language Models Set Reorder Points and Safety Stocks? Industrial Evidence Against ERP and Data-Driven Benchmarks

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Abstract

How should training-free large language model (LLM) replenishment proposals be screened before operational consideration? *InventOR* supplies auditable industrial evidence through deterministic cost, service and trajectory evaluation. For 365 plant-material pairs from three plants, three original generations, seven reference arms and an ERP-input ablation are evaluated on held-out recorded series. Three masked generations and rescaled originals share 338 common pairs, not independent datasets. On the original 346-pair common-feasible cohort, LLM policies cost about 11% less than ERP-derived parameters and 30–31% less than code using ERP planning inputs, but 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile baselines. Mean fill is 0.1–0.4 percentage points below ERP-derived policies; volume-weighted fill is much lower than mean material fill. Plant C and priced shortages reverse the pooled ERP comparison. Static storage checks miss trajectory breaches, while capacity clipping releases sub-MOQ orders. A separately identified offline intervention reuses frozen policies with strict admission, supported by a conditional invariant and finite-state counterexamples, without guaranteeing service or optimality. Masked scoring supports numerical replay, not regeneration of proprietary inference. Source-demand semantics and additive grain remain unverified. The practical finding is comparator-aware screening with

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joint order-admission checks; measured planner benefits and operational deployment remain unevaluated.
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1. Introduction

Manufacturing replenishment requires plant-material reorder points, safety stocks and order quantities. Revising them requires demand/lead-time estimation, inventory theory, minimum order quantity (MOQ) and storage checks, and planner explanations. LLMs with code interpreters are proposed for supply-chain decisions (Fosso Wamba et al., 2023; Fosso Wamba et al., 2024; Jackson et al., 2024; Panja et al., 2026; Song et al., 2026; Trifone et al., 2026), extending data analytics in logistics (G. Wang et al., 2016); their numerical proposals still require independent evaluation.

Three problems limit its use. First, LLM outputs can be unfaithful or fabricated (Ji et al., 2023), so a proposed policy is a candidate, not a decision. Second, superiority claims over ERP parameters are hard to interpret when the LLM and the comparators receive different information. Third, a storage rule that holds when a policy is written down can be violated as stock builds up over time.

The empirical problem is to distinguish an attractive ERP comparison from a proposal worth reviewing. InventOR evaluates training-free, fixed per-material (r, Q) proposals against seven reference arms with explicit input boundaries and trajectory checks. It is not a new optimization algorithm or generator/verifier architecture: recent work already addresses mechanism-aware verification, simulation integration and inventory-policy evolution (Section 3). An enterprise LLM receives a fixed prompt without project-specific training and proposes a policy, justification and escalation flag. Deterministic software parses, gates and

simulates proposals; planner authority is a proposed deployment boundary (Section 2).

We study five questions on real industrial data: what cost, service and inventory the LLM policies achieve relative to ERP-derived parameters, data-driven baselines and textbook code, with explicit input boundaries (RQ1); how their differences relate descriptively to parameter similarity, run-to-run noise, demand segments and shortage prices (RQ2); whether policies respect storage limits along simulated trajectories (RQ3); how stable they are across repeated generations (RQ4); and which plant and demand-regime results might inform proposed review and escalation rules (RQ5).

We contribute (C1) comparator-aware industrial evidence for screening training-free proposals (Sections 2 and 4); (C2) seven reference arms, an ERP-input ablation and trajectory-level MOQ/storage diagnostics (Section 5); (C3) frozen original and masked-policy replay with a matched six-run report, plus separately labeled strict-admission support (Section 6; Supplementary File S2); and (C4) conditional operational implications and testable next steps for planner evaluation (Section 7).

Against ERP-derived parameters the original LLM policies cost about 11% less at mean fill 0.1–0.4 points lower, but cost 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile policies. Code using ERP inputs costs 44–45% more than the LLM, showing the importance of input treatment. Median policy ratios resemble textbook code, and the six-run matched report preserves the ranking. Static-compliant policies exceed storage limits on unconstrained trajectories for 53–66% of pairs, which motivates trajectory-level gating rather than a claim of operational feasibility.

2. Problem Setting and Workflow

2.1. The decision problem

The decision supported is the periodic (re)setting of a reorder point, a safety stock and an order quantity for each plant-material pair held in an ERP system. It is a partly structured operational-control decision (Gorry & Scott Morton, 1971): the calculation is textbook, but the choice of estimator, the treatment of intermittent demand, the plausibility of recorded lead times and the handling of constraints require judgement that is applied inconsistently when thousands of materials compete for a few analysts' time. The decision-maker is the material planner; hard constraints are the minimum order quantity (MOQ) and a physical storage limit; and feedback is slow, because a changed parameter shows its effect only in later demand. The scope is one pair at a time (single item, single echelon, continuous review, lost sales, (r, Q) family); order release, supplier selection, forecasting-model development and multi-echelon networks are out of scope. The LLM proposes candidates; it does not place orders, make deployment decisions or run the study's deterministic held-out evaluation replay.

2.2. The InventOR workflow and its level of automation

> **Definition.** *InventOR is a human-reviewed planning workflow that, for one plant-material pair at a time, (i) obtains from a large language model, configured only by a written prompt, a candidate (r, Q) policy with a justification and an escalation flag; (ii) verifies the candidate deterministically against explicit constraints; (iii) evaluates it by counterfactual simulation against reference policies on held-out demand; and (iv) presents the result to a planner, who alone decides whether ERP parameters are changed.*

In the levels of automation of Parasuraman et al. (2000), InventOR operates at a low-to-intermediate level: it suggests one alternative and leaves execution to the human. Accepting suggestions invites complacency and automation bias (Parasuraman & Manzey, 2010), which the review rules of Section 4.3 are meant to counter.

2.3. Requirements on the workflow

The requirements follow from the known unreliability of LLM outputs (Ji et al., 2023), guidance on human–AI interaction and accountability (Amershi et al., 2019; Raji et al., 2020), run-to-run nondeterminism of LLM inference (Ouyang et al., 2025) and the constraints of inventory control. Table 1 ties each to a design feature (Section 4) and an evaluation (Section 6).

Table 1: Requirements on the workflow, the features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.10
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.8
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

3. Related Work and Positioning

Industrial inventory evidence and validation.. Bai et al. (2012) study a practical batch-ordering policy under incomplete demand knowledge; Fu and Chien (2019) link intermittent-demand forecasting to an electronics-distribution application. These motivate implementable comparators and explicit information boundaries, not the assumption that better forecasts

ensure better policies. Lau et al. (2008) compare inventory policies under simulated operating conditions, supporting conditional cost/service interpretation. InventOR likewise distinguishes estimation, static feasibility and realized replay outcomes; its recorded-series, lost-sales replay is not a replication of those experiments.

Human–AI decision making in planning.. Decision support aims to improve human judgement, not replace it (Gorry & Scott Morton, 1971; Shim et al., 2002; Sprague, 1980); suggestion scope and verification influence appropriate reliance (Parasuraman & Manzey, 2010; Parasuraman et al., 2000). Using a retail-company case and simulation, X. Wang et al. (2024) document algorithm aversion in safety-stock decisions during disruptions and report better algorithmic performance in that setting. People also discount algorithms after seeing them err (Dietvorst et al., 2015). In a German retail-logistics case, Loske and Klumpp (2021) report efficiency advantages for two depots fully using AI routing under their DEA analyses. Neither finding establishes benefits from InventOR, which retains planner authority.

LLMs in operations and supply-chain decisions.. Wu (2026) compares automated and human-specified LLM-assisted inventory optimization under disruptions. Zabraoui et al. (2026) address conversational inventory analytics with an EOQ module. Y. Wang et al. (2025) separate candidate generation from scheduling evaluation; El Baz et al. (2026) combine agentic warehouse optimization and simulation. Yan and Ji (2026) describe mechanism-aware multi-agent scheduling with generation–verification, and SimGPT integrates Simio with LLMs across simulation generation, execution and analysis (Dehghani, 2026). InvEvolve's arXiv v4 preprint describes reinforcement-learning-trained LLMs and online inventory-policy evolution with confidence-based certification (Huang et al., 2026). These are preview/abstract-supported descriptions, not independent verification of their methods or guarantees. InventOR instead evaluates fixed prompt-configured proposals

without project-specific training; it does not benchmark or claim superiority over these systems. Reviews map generative AI opportunities in supply chains (Fosso Wamba et al., 2023; Fosso Wamba et al., 2024; Jackson et al., 2024; L. Li et al., 2024; Panja et al., 2026; Song et al., 2026) and emphasize human roles, data quality and governance (Cannas et al., 2024; Trifone et al., 2026; Venkatesh et al., 2024). Examples include fine-tuned digital-manufacturing optimization (Amarasinghe et al., 2026), learned order timing and quantity (Liu et al., 2026), and managerial decision biases (Chen et al., 2025); these examples do not establish field-wide prevalence. Optimization modeling/code generation (Ahmaditeshnizi et al., 2024; Huang et al., 2025; B. Zhang et al., 2025) differs from OptiGuide's natural-language explanation and what-if interaction with solver outcomes (B. Li et al., 2023), and from multi-agent inventory control (Jannelli et al., 2025; Quan & Liu, 2024). InventoryBench evaluates OR algorithms, LLM agents and humans on more than 1,000 instances, reporting benefits from OR-augmented LLMs (Baek et al., 2026). Our narrower question concerns fixed policies on an industrial extract. In a controlled retail-forecasting experiment with 123 human forecasters and five LLMs, Abolghasemi et al. (2025) find that the tested LLMs do not consistently outperform humans. Abbaszadeh Nakhost et al. (2025) examine LLM-based spare-parts demand forecasting. We test policy variability directly (Section 6.8), rather than infer it from forecasting results.

Data-driven inventory control without LLMs.. Learning-based replenishment includes non-stationary demand policies (Ren et al., 2024), leading indicators (Sagaert & Kourentzes, 2025), deep reinforcement learning for omnichannel and multi-echelon systems (Kolyaei et al., 2025; Ma, 2026; Stranieri et al., 2024), contextual robust optimization (Q. Wang et al., 2026) and forecasting-linked reinforcement learning (Schett et al., 2026). For finite-horizon perishable inventory with non-stationary demand and zero lead time, Gulecyuz et al. (2025)

compare a computationally efficient heuristic with optimal stochastic dynamic programming. Gabellini et al. (2026) evaluate joint replenishment, safety-stock and safety-lead-time decisions on real automotive-company data. Forecasting research links forecasts to safety stock (Beutel & Minner, 2012), combined-error distributions (Barrow & Kourentzes, 2016) and inventory costs (Kourentzes et al., 2020). Model-based deep reinforcement learning addresses new products with short histories (Demizu et al., 2023); contextual inventory control is studied by Batsis and Samothrakis (2024). TFT-MPIR generates multi-period orders from historical/contextual inputs (Guo et al., 2025); graph recurrent networks address store-SKU replenishment (Ma, 2024); RevIN-TSMixer connects forecasts to continuous-review inventory costs (Nguyen et al., 2026). InventOR evaluates a fixed-prompt pretrained generator, not superiority over these trained methods.

Intermittent and spare-parts demand.. Manufacturing histories are often intermittent or lumpy. Work began with the bias of Croston's estimator and its stock-control performance (Syntetos & Boylan, 2001, 2006) and continues with neural (Kourentzes, 2013), bootstrapping (Hasni et al., 2019), obsolescence-aware (Babai et al., 2019), temporal-hierarchy (Sanguri et al., 2023) and state-space (Svetunkov & Boylan, 2023) forecasts, order-up-to control (Rostami-Tabar & Disney, 2023), safety-stock placement (Yazdani & Aouam, 2023) and lead-time forecasting (T. Zhang & Xu, 2026), all of which bear on the inputs of the policies we compare. The wider literature adds specialized estimators and classification schemes (Babai et al., 2026; Croston, 1972; Prak et al., 2021; Syntetos & Boylan, 2005; Willemain et al., 2004), spare-parts forecasting (Goldsmith & Sachs, 2025) and multi-echelon spare-parts models (Caserta & D'Angelo, 2026). InventOR is not a new inventory model and does not claim to beat trained or optimized methods; its (r, Q) policy is the standard workhorse of continuous-review control (Hadley & Whitin, 1963; Porteus, 2002; Silver et al., 2017).

4. The InventOR Workflow

4.1. Architecture and components

Following classical decision-support components (Shim et al., 2002; Sprague, 1980), InventOR comprises *data* (material-scoped histories, costs, constraints and ERP values; DR4); *knowledge* (the frozen task, prohibited methods, rules and schema in Supplementary File S1; DR1, DR5); *models* (a prompt-configured enterprise LLM/code-interpreter generator and a separate deterministic parser, gate and held-out simulator; DR1–DR3); *dialog* (policy, justification, flag and comparison for proposed planner review; DR6); and *governance* (input hashes, timestamps, sessions, raw responses and repeats; DR5). The generator is non-inspectable. The evaluated dialog is a per-pair output record; no interactive interface was built or evaluated.

4.2. Operation in four decision steps

For one material, InventOR uses four decision steps after input preparation; Figure 1 includes preparation as a fifth box.

Step 1. Generate. The enterprise LLM/code interpreter receives one material's history and cost, constraint, service-target and ERP parameters (Section 5.2). The written prompt is the only configuration; weights remain unchanged (Sahoo et al., 2024). It requests transparent statistics, a structured policy, justification and escalation flag, and gives an EOQ-related quantity rule with MOQ/storage bounds. It does not prescribe the normal-approximation safety-stock formula used by comparators. All scored artifacts resolve to fixed (r, Q) policies; permitted family labels are not separate evaluated methods.

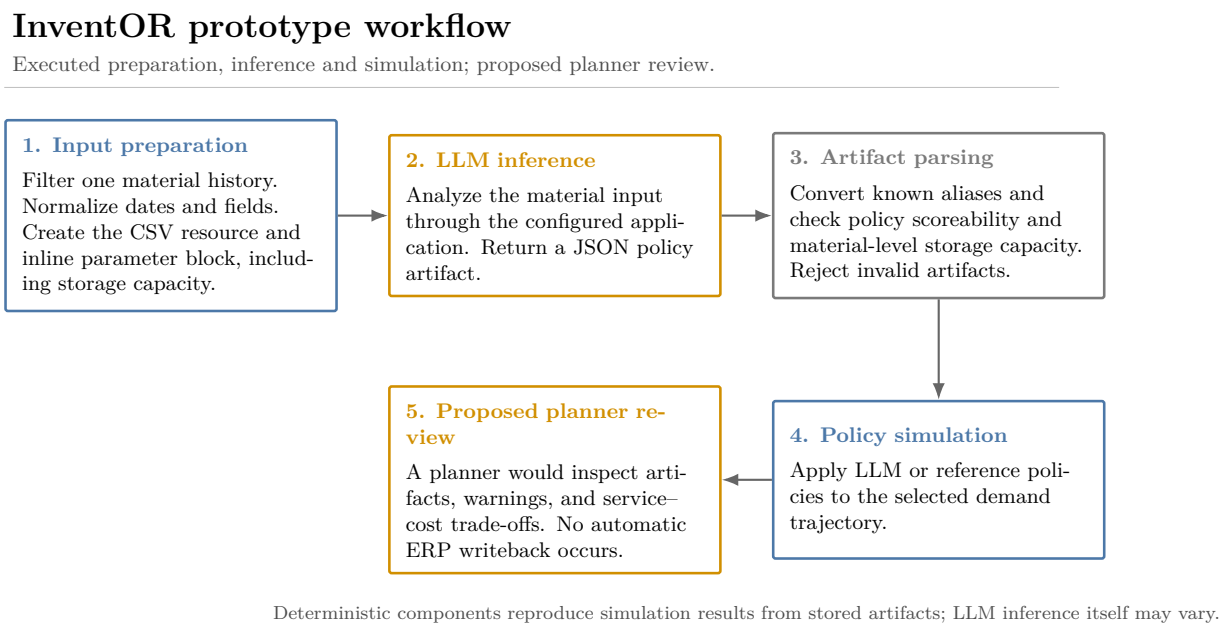


Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, and deterministic software parses, checks and simulates it. Planner review is a proposed deployment step, not an evaluated activity.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material's demand observed *after* the data cutoff, a held-out evaluation in the sense of out-of-sample forecast testing (Hewamalage et al., 2023; Tashman, 2000), together with the comparator policies, producing holding and ordering cost, fill rate and stockout days.

Step 4. Review (proposed deployment step). A planner would inspect the policy, justification, escalation flag and simulated comparison and would retain sole authority over ERP changes. Recording the planner's decision, review time and comments

is proposed future functionality; neither this logging nor planner outcomes was evaluated in the present study.

The LLM is a *candidate generator* whose output is gated; it does not run the study's deterministic held-out evaluation, place orders or decide ERP changes. The prompt prohibits Monte Carlo simulation, distribution fitting, machine-learning libraries and specialized intermittent-demand estimators, so that the policy rests on transparent statistics.

4.3. Trust, review and reject rules

We propose three rules for planners, in line with guidelines for human–AI interaction and AI auditing (Amershi et al., 2019; Raji et al., 2020). The rules and the 0.5-point threshold are design proposals not evaluated with planners; the trust rule would be met by a minority of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346).

Trust (consider after review) only after checking both order-level MOQ and storage compliance, as well as a simulated cost no higher than the best history-only reference (Croston/SBA or empirical quantile) with fill not more than about 0.5 points lower. Capacity-clipped replay alone cannot establish the joint compliance needed to apply this rule.

Escalate when the LLM raises an escalation flag, the material is lumpy or has strongly seasonal or trending demand, the plant or calendar data were repaired, or the policy differs strongly between repeated generations.

Reject when the proposal fails the gate after the retry budget, exceeds the storage limit in the hard-limit replay, or is dominated (higher cost and lower fill) by a history-only reference.

4.4. Provenance and reproducibility of the configuration

The system prompt, deployed unchanged, defines the task, forbidden methods, capacity rules and output schema (Supplementary File S1). Calls log input SHA-256, UTC time, ses-

sion identifier and raw response. The platform does not expose model identifier or sampling settings; Anthropic Claude Sonnet 4.6 with a code interpreter is operator-declared. Frozen policies support numerical replay, not regeneration of proprietary inference. Three repeated generations examine within-panel variation (Section 6.8).

5. Application and Evaluation Design

5.1. Evaluation approach

The evaluation addresses Table 1 through hard-limit replay with working-day receipts and no shortage penalty, plus sensitivities. This ex post simulation replays a processed operational-extract series; its source-demand definition and additive grain have not been validated. There are no live users. The history-only, SLT-informed and ERP-floor OR arms implement textbook EOQ and normal-approximation policies with different ERP-input treatments, not exact execution of a prescribed prompt formula. Outcomes characterize conditional policy cost, service and constraint failures, not usability or planner benefits.

5.2. Industrial application and data

The data come from a large manufacturer, and the evaluation is retrospective. The dataset contains 233,078 daily records for 411 plant-material pairs from three plants (A, B and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed to 27 August 2026. Eligibility requires at least 20 working-day observations before the cutoff and 40 on or after it within the evaluation horizon. Of the 411 pairs, 365 are eligible for scoring (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19 pairs (all in Plant A), at least one of the SAP-derived or SLT-informed reference policies cannot be projected to satisfy MOQ and the static $SS + Q$ capacity rule while retaining its safety stock; these are excluded from outcome comparison, leaving a *common-feasible cohort* of

346 pairs (215, 65 and 66). This is a comparator-specific exclusion, not proof of general policy infeasibility. Demand is concentrated (the five largest materials account for 40.3% of demand). The source working-day flag of Plant C is all zero; we reconstruct it from the same-date majority of Plants A and B and test this repair in Section 6.10. Data quality is a known threat to decision support (Watts et al., 2009) and inventory records (DeHoratius & Raman, 2008; Kang & Gershwin, 2005; Mersereau, 2013).

The extract has 220,017 distinct plant-material/date groups. A masked-panel diagnostic finds 12,036 repeated groups across 62 of the full panel's 411 pairs: daily quantities repeat unchanged while context varies. Frozen scoring sums signed same-date demand before taking its absolute value, averages positive lead times and retains the last row's other attributes; stock is not summed. Without a source dictionary and extract-grain reconciliation, these are reproducible processing choices, not verified aggregation of distinct demand events. The implemented additive grain, sign handling, context selection and stock timing require source validation. The implementation is in `inventory_llm/core.py`: `load_grouped_rows`, `demand`, `compute_stats` and `eligible_item_keys`.

Each pair is sent as a material-scoped CSV with daily demand, recorded lead times, a working-day flag, the minimum order quantity, unit cost, holding-cost rate, ordering cost, service-level target, the storage limit, and two *ERP planning parameters* (current safety stock and safety lead time), which are withheld in the ablation arm. The CSV also carries supplier and lane context fields that the prompt does not use. A masked version of the panel is released for replication (Section 6.9); there the plants are labelled PL1, PL2 and PL3 (corresponding to A, B and C).

5.3. Reference arms and information content

ERP safety stock and safety lead time embed planner judgement and are alternative ways of buffering unreliability (van Kampen et al., 2010), and planned lead times can drift upward (Thürer et al., 2023). The LLM and the classical comparators therefore do not necessarily use the same information, and we construct arms that make it explicit (Table 2).

Table 2: Comparison arms and their information. Policy quantities satisfy MOQ and are projected to the storage limit when possible; capacity-clipped orders in replay can fall below MOQ.

Arm	Parameters from	ERP inputs used	Purpose
SAP-derived	ERP safety stock and safety lead time; EOQ order quantity	Both	ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time	OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both	OR arm matched to the LLM's full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None	ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None	Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None	Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None	Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both	Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to "not supplied"	None	Ablation: what the LLM does when it cannot anchor on ERP

Three further arms use no ERP planning inputs. *Croston/SBA OR* uses bias-corrected Croston demand (Croston, 1972; Syntetos & Boylan, 2005), smoothing 0.1, and root mean squared one-step forecast error as demand spread, a study-specific heuristic rather than a validated lead-time-demand distribution. *Empirical-quantile OR* uses the service-target

quantile of rolling demand over rounded mean lead time. *Simulation-tuned* (r, Q) searches reserve factors $-0.5, -0.25, \ldots, 3.5$ and quantity multiples $0.5, 0.75, 1, 1.5, 2$ (up to 85 candidates), selecting the cheapest meeting the service target, or the best fill if none meets it. Statistics use all pre-cutoff data; tuning replays the last 250 working days without capacity clipping, initialized at each candidate's $R + Q$ with an empty pipeline. This is in-sample tuning, not nested validation; evaluation instead uses shared observed opening stock and capacity clipping. No post-cutoff demand informs selection. Deep reinforcement learning (Kolyaei et al., 2025; Stranieri et al., 2024) is not benchmarked.

5.4. Simulation, capacity replay and metrics

Each pair is replayed as a single-item, single-echelon (r, Q) system (Axsäter, 2015; Hadley & Whitin, 1963; Porteus, 2002; Silver et al., 2017; Zipkin, 2000) with lost sales (Bijvank & Vis, 2011), discrete working-day review, shared observed opening stock, an empty pipeline and deterministic mean-lead-time arrivals (Appendix A). This is simulation, not reconstruction of actual orders (Grasso & Taylor, 1984). Fill and shortage cost are conditional on the processed recorded series: whether it represents availability-independent requests or availability-influenced fulfilled issues is unverified; no latent-demand or historical lost-sales reconstruction was performed. Policy quantities satisfy MOQ (Heese, 2025), but capacity-clipped orders (X. Zhang et al., 2024) may not. Metrics are mean material-horizon holding-plus-ordering cost, mean and volume-weighted fill, and stockout days. Paired 5,000-resample percentile-bootstrap 95% intervals (seed 20261006) describe this cohort; dependence within three plants precludes population-level inference.

A static check $SS + Q \leq$ `max_storage_units` does not show that storage limits hold as inventory evolves, because the reorder point includes lead-time demand and inventory on order can exceed the limit on arrival. We therefore record the post-receipt on-hand stock for

every policy and day and report the share of pairs and days above the limit (the *unconstrained* simulation), and we *replay* every policy with each order truncated to the storage limit minus the current inventory position (the *hard-limit replay*), which is our primary evaluation.

5.5. Repeated runs and sensitivities

To examine run-to-run variation, the entire cohort is generated three times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1 serves as the reference for tabulated results unless stated otherwise; runs 2 and 3 provide replicates. Sensitivity analyses vary the working-day reconstruction (majority, any-plant, strict-all, exclusion of Plant C and calendar-day arrivals) and the shortage penalty (zero, one and five times unit cost).

6. Results

6.1. Completion and policy characteristics

All four original-data runs (three full-input and one ablation, Section 6.4) return an (r, Q) policy satisfying the numerical and static storage gate for all 365 pairs after at most three attempts per pair. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm holds no safety stock at all.

6.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, all results use the hard-limit replay on the 346-pair common-feasible cohort with working-day receipts and no shortage penalty (Section 6.5 gives the unconstrained simulation). Table 3 compares all arms; its last columns give the LLM cost relative to each arm in the three runs (positive: the LLM costs more). In run 1 the LLM

policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR (mean 2,928 versus 3,305 and 4,211), at a mean fill 0.18 points lower than SAP-derived (98.11% versus 98.30%; interval −0.54 to +0.10). The mean cost difference against SAP-derived is −377 (95% interval −651 to −154); the LLM is cheaper in 241 of 346 pairs and, in 226, simultaneously no more expensive and no worse in fill. It incurs more stockout days (910 versus 804). The volume-weighted (aggregate) fill rate is only 78.7–79.3% for every arm except the simulation-tuned policy (84.8%), against a mean material-level fill of about 98%: with lost sales unpriced, service on high-volume materials is low under all policies, so the cost ranking is not a comparison at equal service.

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same mean number of orders per material over its evaluation horizon (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.10 shows that once shortages carry a cost this trade no longer favours the LLM.

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is mean material-horizon holding-plus-ordering cost; inventory is in units. Orders are mean counts per material over its evaluation horizon; stockout days are summed. The last column gives LLM cost relative to the arm in runs 1/2/3 (positive: LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	−11.4 / −11.3 / −11.4
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	−30.5 / −30.4 / −30.5
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	−31.1 / −31.0 / −31.1
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	+5.9 / +6.0 / +5.9
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	+8.2 / +8.4 / +8.3
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	+17.7 / +17.8 / +17.7
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	−10.4 / −10.3 / −10.3‡
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	−8.8† (vs SAP)

† Other comparisons of the ablation are in Section 6.4. ‡ Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.10).

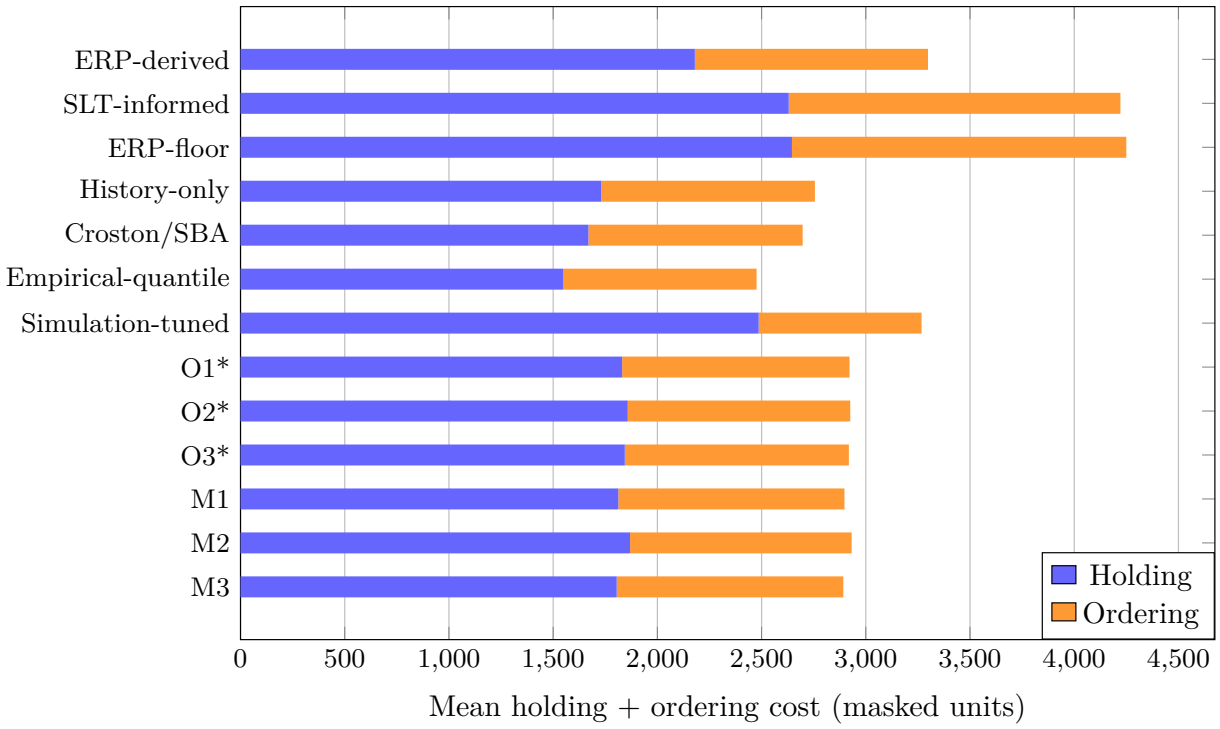


Figure 2: Six-run cost decomposition on the same 338 feasible masked-panel pairs (zero shortage penalty). O1*–O3* are rescaled frozen original policies, not fresh inference; M1–M3 are new masked generations. Comparators are identical across replays. Reciprocal quantity/price scaling preserves economic products before rounding; these costs do not demonstrate monetary confidentiality. Table 3 uses the original cohort/protocol.

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mostly measures how heavily ERP inputs over-buffer.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with fill differences of −0.03, −0.25 and +0.01 points whose intervals include zero, and is cheaper in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97 to +362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489,

fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM's advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean fill of 96.05% and 1,973 stockout days, because it is tuned on the last 250 pre-cutoff working days. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher mean fill, but the interval includes zero (−2,527 to +857); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.10).

6.4. LLM without ERP planning parameters (ablation)

This ablation blanks ERP safety stock and safety lead time, marking them "not supplied", with the same prompt and other inputs. All 365 original pairs returned a static-compliant policy. Median parameter ratios to run 1 are 1.00, but reorder points differ by more than 10% in 54.6% of pairs, against 48.3% and 48.0% between full-input runs. Hard-limit cost is 3,014 at mean fill 98.06%: 8.8% below SAP-derived (interval −566 to −61), but 9.0% above History-only (interval +134 to +387), 11.4% above Croston/SBA and 21.1% above empirical-quantile. Relative to SAP-derived, costs are −10.8%, −21.9% and +41.6% in Plants A/B/C. The Plant C reversal therefore persists without these fields; one stochastic ablation does not isolate their causal effect or prove that the LLM ignores them. Costs, MOQ, service target and capacity remain available to all arms.

6.5. Storage limits along simulated trajectories

All 365 original LLM policies satisfy $SS + Q \leq$ `max_storage_units`, yet unconstrained post-receipt stock exceeds the limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for SLT-informed

OR. The static rule does not bound pipeline accumulation and is insufficient for trajectory feasibility. In the hard-limit replay, clipping keeps every arm within one unit of the limit (integer truncation of inventory position), but clipped orders may fall below the MOQ. Enforcing capacity raises LLM mean cost from 2,850 to 2,928 and SAP-derived from 3,184 to 3,305 without changing the ranking.

6.6. Plants and demand regimes

The average hides a strong plant effect. In original run 1 the LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C (cheaper in 8 of 66); the Plant C loss is similar in the other full runs (+32.2%, +31.5%) and persists under every working-day reconstruction in which Plant C can be scored (Section 6.10). Figure 3 separately shows all six runs on the matched masked-scale cohort. Against the data-driven baselines the original run 1 LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only OR, 13.1%, 2.3% and 6.2% more than the Croston/SBA OR, and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR in Plants A, B and C. The SLT-informed and history-only ORs coincide in Plant C because ERP safety lead time is zero for all its pairs.

By demand regime, a study-specific high-variability proxy (pre-cutoff working-day demand coefficient of variation CV ≥ 1) identifies 33 materials in the 346-pair original cohort. Labelled "lumpy" in the evidence code, this is not the conventional ADI/CV^2 taxonomy. They show the largest reduction against SAP-derived (-25.4% in run 1), but the history-only and Croston/SBA ORs cost about 9–10% less than the LLM there; with 33 materials this is indicative only.

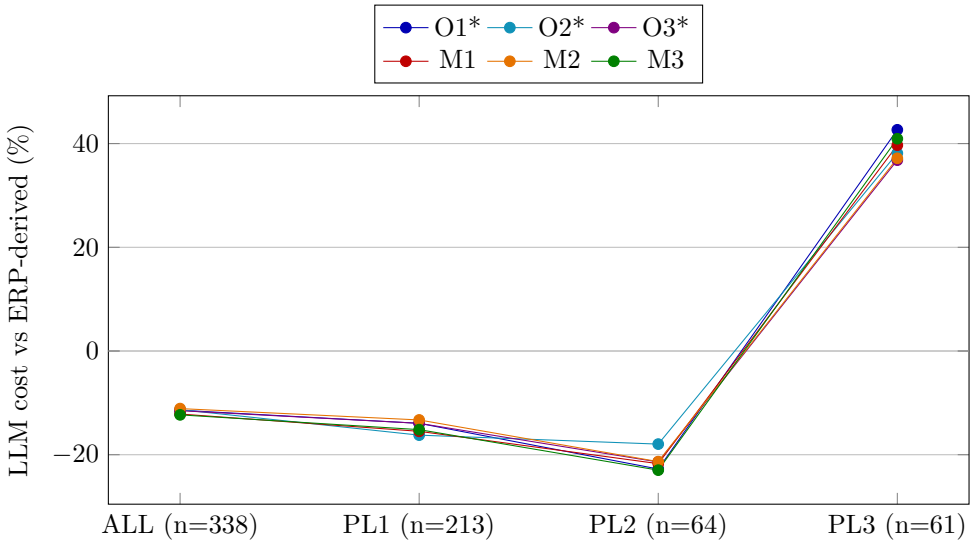


Figure 3: Plant heterogeneity across all six runs on a shared masked-panel cohort. Positive values mean the LLM costs more than ERP-derived parameters. O1*–O3* denote rescaled frozen original policies; M1–M3 denote fresh masked generations. PL1/PL2/PL3 correspond to Plants A/B/C. Lines connect descriptive group summaries, not time points or uncertainty intervals.

6.7. Descriptive comparison with formula-based code

The deterministic History-only OR uses textbook EOQ with an MOQ floor and normal-approximation safety stock (Appendix A); the SLT-informed and ERP-floor ORs additionally use ERP inputs available to the LLM (Section 5). The historical prompt constrains the economic quantity and permits several policy families; it does not prescribe these comparator formulas in full. This is a descriptive comparison with reference code, not a test of compliance with a prescribed normal-approximation recipe.

The LLM resembles the formula-based policies.. The median smoothed ratio $(x_{\text{LLM}} + 1)/(x_{\text{code}} + 1)$ is 1.00 for order quantity (medians 2,810 and 2,828 units) and safety stock (712 and 718, against an ERP median of 0); LLM parameters are averaged over the three original runs first. Here "code" denotes History-only OR. The corresponding reorder-point ratio is 1.09, including across the segments in Table 4, where LLM cost is 4–9% higher. These summaries indicate similarity and a reorder-point uplift, not item-level proof that the LLM executes the formulas faithfully.

Input treatment changes the ranking against code.. Code that receives the same ERP inputs as the LLM costs 4,211 (SLT-informed OR) and 4,248 (ERP-floor OR) per pair against 2,928 for the LLM in run 1, i.e. 44–45% more than the LLM, because the ERP safety lead times inflate the textbook reorder point. Code that ignores the ERP planning inputs costs 2,764; the LLM costs 5.9% more than this code. The LLM resembles the History-only code rather than the ERP-inflated code, which is consistent with the ablation in Section 6.4.

Run-to-run noise.. The within-pair cost range across the three original runs, divided by that pair's mean and averaged across pairs, is 10.5%; this is not a standard deviation. The runs disagree on beating History-only code for 34.1% of pairs. The LLM mean is cheaper than History-only for 17.1% of pairs and SAP-derived for 76.3%. Selecting each pair's best run with hindsight gives mean cost 2,765 (History-only: 2,764); selecting the worst gives 3,146. Such selection is not an implementable out-of-sample policy.

Plant C and segments.. Table 4 shows that the LLM advantage over SAP-derived parameters disappears in storage-bound pairs (+4.9% against −14.9% elsewhere) and reverses in Plant C (+33.3%); it does not vary monotonically with demand variability. Plant C is not an LLM-specific failure: the History-only code is also 28% more costly than SAP-derived parameters there, so the ERP parameters are leaner than any formula-based policy in that plant when lost sales are not charged. Plant C also has the least regular demand (median coefficient of variation of daily demand 0.61, lead-time coefficient of variation 0.78, against 0.51 and 0.63 over all pairs).

6.8. Run-to-run stability

Original aggregate costs relative to SAP-derived are −11.4%, −11.3% and −11.4%. Figure 4 extends the ranking comparison to all six runs on shared masked-panel pairs. Individual

Table 4: Where the LLM differs from code. Original-scale mean cost per pair in the hard-limit replay with zero shortage penalty (LLM: mean of three original runs; History-only: textbook code; SAP: ERP-derived), relative costs (%), and the median smoothed reorder-point ratio $(R_{\text{LLM}} + 1)/(R_{\text{code}} + 1)$. Storage-bound pairs require downward projection because textbook safety stock plus order quantity exceeds capacity.

Segment	n	LLM	History-only	SAP	vs History-only (%)	vs SAP (%)	ROP ratio
All pairs	346	2,929	2,764	3,305	+6.0	-11.4	1.09
Plant A	215	2,370	2,231	2,782	+6.2	-14.8	1.08
Plant B	65	5,050	4,746	6,360	+6.4	-20.6	1.10
Plant C	66	2,662	2,549	1,997	+4.4	+33.3	1.11
Demand CV, lowest third	115	2,249	2,092	2,707	+7.5	-16.9	1.09
Demand CV, middle third	115	3,872	3,659	4,152	+5.8	-6.8	1.10
Demand CV, highest third	116	2,669	2,544	3,057	+4.9	-12.7	1.08
ERP safety stock zero	188	2,946	2,741	3,215	+7.5	-8.4	1.09
ERP safety stock positive	158	2,909	2,792	3,411	+4.2	-14.7	1.09
Storage-bound	145	1,491	1,408	1,422	+5.9	+4.9	1.10
Not storage-bound	201	3,967	3,743	4,663	+6.0	-14.9	1.08

policies still vary: original runs 2 and 3 have identical triples for only 36 of 365 pairs. Similar aggregate means do not establish item-level repeatability (Ouyang et al., 2025).

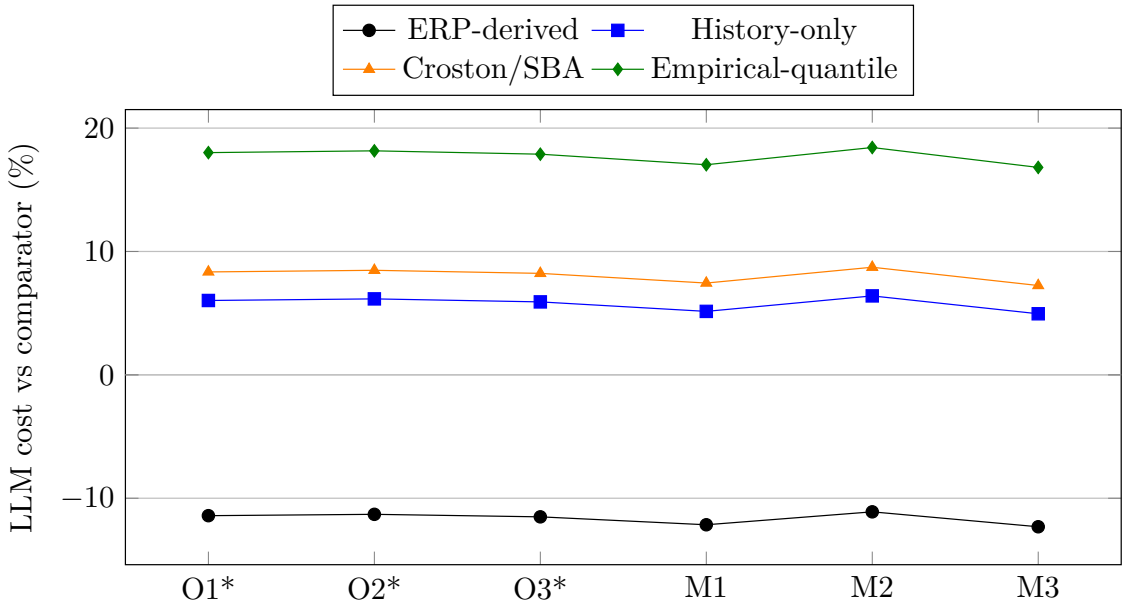


Figure 4: Ranking stability across six replays on 338 shared masked-panel pairs, zero shortage penalty. Negative values favour the LLM. O1*–O3* are rescaled original policies; M1–M3 are masked generations. The consistent signs support aggregate ranking stability on this panel, not independent replication across datasets or stability of individual policies.

6.9. Replication on the masked release

The raw data cannot be shared; an existing masked panel supports replay (see Data and Code Availability). Quantities are multiplied by secret per-pair factors and unit prices divided by those factors, preserving economic products before rounding. Identi-

fiers are pseudonyms; dates, lead times and plant structure remain unchanged. This is pseudonymized, not demonstrated anonymous; unchanged dates and preserved economic products could still allow linkage to the source. Masking and employment establish neither redistribution rights nor permission. We repeated generation three times on this panel with the unchanged prompt (m1–m3). Run m1 produced 364 and m2/m3 all 365 policies (one pair exhausted retries in m1); 340, 342 and 342 were scorable, and 5, 4 and 4 invalid. Table 5 requires all seven comparators; SLT-informed and ERP-floor costs appear in Figure 2.

Table 5: Six-run comparison on 338 pairs feasible for all runs and seven comparators. Cost is mean holding-plus-ordering cost of masked-panel replay; fills and cost differences are percentages. O1*–O3* are rescaled frozen originals, M1–M3 fresh masked generations. Cohort inclusion, generation and rounding differ from Table 3; economic products survive reciprocal scaling. Six runs are not independent datasets.

Run	Cost	Mean fill	Aggr. fill	vs ERP	vs History	vs SBA	vs Empirical	vs Tuned
O1*	2,922	98.34	76.44	−11.4	+6.0	+8.3	+18.0	−10.6
O2*	2,925	98.11	76.27	−11.3	+6.2	+8.5	+18.2	−10.5
O3*	2,919	98.37	76.46	−11.5	+5.9	+8.2	+17.9	−10.7
M1	2,898	98.33	76.34	−12.2	+5.1	+7.4	+17.0	−11.3
M2	2,932	98.37	76.38	−11.1	+6.4	+8.7	+18.4	−10.3
M3	2,892	98.37	76.29	−12.3	+4.9	+7.2	+16.8	−11.5

On each masked run's own feasible cohort, the LLM is 11–12% cheaper than ERP-derived parameters and 5–6%, 7–9% and 16–18% costlier than History-only, Croston/SBA and empirical-quantile code. Matched costs relative to rescaled original run 1 are 0.991, 1.004 and 0.991, not comparisons with confidential unscaled means. Table 5 and Figures 2–4 use the same 338 pairs; Figure 3 confirms the PL3 reversal. Quantity scaling changes volume-fill weights; generation, cohort inclusion and integer rounding can change cost summaries despite preserved economic products. Repeated processing does not validate source grain or establish monetary confidentiality.

6.10. Sensitivity analyses

Plant C calendar.. Plant C's working-day flag required reconstruction, so we re-scored all policies under four alternatives (any-plant, strict-all, exclusion of Plant C, calendar-day

arrivals). Where Plant C can be scored, the LLM remains 31–35% more expensive than SAP-derived there, so the Plant C result is not an artefact of the repair.

Shortage penalty.. Rescoring the original fixed trajectories at one and five times unit cost makes the simulation-tuned policy about 33% cheaper than SAP-derived; LLM run 1 costs 2.4% more than SAP-derived at both multiples. These are accounting sensitivities, not policy optimization at each penalty. Table 6 retains the original-scale analysis; the public six-run evidence also records shortage exposure for masked-scale rescoring.

Break-even shortage price.. Table 6 re-scores the same replays for a lost-sales penalty between zero and one unit cost per unit. The LLM is cheaper than SAP-derived parameters only while the penalty stays below 0.9% of the unit cost, and cheaper than the History-only and empirical-quantile code only above 1.5% and 2.4%, by about 1% or less. It is cheaper than the SLT-informed and ERP-floor ORs at every penalty and cheaper than the simulation-tuned policy only below 0.06%. From a penalty of 1% of the unit cost, the SAP-derived, History-only, Croston/SBA, empirical-quantile and LLM policies lie within 4% of each other, while the simulation-tuned policy is 28–33% cheaper than SAP-derived parameters. The ranking between the LLM and the formulas is thus decided by a shortage price that the primary basis sets to zero, which is why it is reported as a sensitivity and not as a headline result.

Table 6: Break-even shortage price. Mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay when each lost unit is charged at the given multiple of the unit cost, and the break-even multiple at which the LLM and the arm have equal mean cost (common-feasible cohort, $n = 346$).

Arm	0	0.01	0.1	1	Break-even multiple of unit cost
SAP-derived	3,305	19,613	166,388	1,634,136	0.0089; LLM cheaper below
History-only OR	2,764	19,609	171,207	1,687,191	0.0145; LLM cheaper above
Croston/SBA OR	2,705	19,427	169,927	1,674,925	none; LLM costlier on the grid
Empirical-quantile OR	2,489	19,402	171,618	1,693,784	0.0242; LLM cheaper above
Simulation-tuned	3,266	14,174	112,344	1,094,043	0.0006; LLM cheaper below
LLM (mean of three runs)	2,929	19,660	170,238	1,676,014	–

Secondary strict-admission validation.. Supplementary File S2 separately replays the same 338 pairs after reproducing frozen clipping results. Fixed policies use floored physical headroom and defer releases below MOQ. Strict replay has no sub-MOQ releases or capacity breaches; M1 mean fill falls from 98.326% to 98.323%, with 48 deferrals, and Plant C cost remains above ERP-derived (2,599 versus 1,845 masked units). A conditional admission invariant and independent finite-state/counterexample tests support the simulator property, not service or optimality. Initialization is hypothetical; opening excess is retained and excluded from the invariant, not descriptive summaries (none on this cohort). Both modes report cost, mean/volume fill and adverse outcomes. Other local diagnostics are exploratory, outside S2 and nonessential; separately retuned arms do not match held-out service.

7. Discussion, Limitations and Research Perspectives

What the evidence shows.. Cost/service/trajectory screening distinguishes ERP-relative savings from competitive proposals. All original (r, Q) proposals passed the static gate; masked generations include missing and invalid outputs. Policy similarity and run variation characterize outcomes, not internal calculations. S2 makes a narrow admission-rule intervention inspectable without rewriting the original experiment. Neither that intervention nor input matching establishes equal-service superiority.

Implications for practice.. (1) Benchmark against data-driven code, not ERP alone; input treatment matters. (2) Price shortages before interpreting cost rankings (Section 6.10). (3) Verify storage and MOQ jointly: static compliance and capacity clipping do not ensure both. (4) Expect plant differences; define review/escalation rules (Parasuraman & Manzey, 2010). (5) Version prompts, retain provenance and repeat generations. (6) No project-specific training was needed, but measured runtime, model fees, engineering effort and planner time

were not collected; low configuration burden is not evidence of fast or cheap deployment.

Limitations.. The evidence comes from one manufacturer's three-plant panel, which the masked generations reuse; source-demand and data-grain validation remains outstanding: repeated context rows leave the additive grain unverified, and the recorded series need not reflect availability-independent requests. The model identity is operator-declared, planner outcomes were not measured, and fixed-policy comparisons are not equal-service; separately retuned arms do not match held-out service. Policies are single-item (r, Q), and trained neural or reinforcement-learning baselines are left to future work.

Research perspectives.. (1) *Constraint-aware hybrids:* test mechanism-aware generation with independent joint-feasibility checks, motivated by Yan–Ji and SimGPT, without transferring performance claims. Policy aggregation is untested here, distinct from spare-parts forecasting (Abbaszadeh Nakhost et al., 2025); LLM review of data-driven policies (Wasserkrug et al., 2025) also needs evaluation. (2) *Matched comparisons:* compare InvEvolve, trained inventory-control frameworks (Gabellini et al., 2026), reinforcement learning (Kolyaei et al., 2025; Stranieri et al., 2024) and richer (s, S) policies (Babai et al., 2026; Prak et al., 2021) at matched information, service and measured setup/inference/review budgets. (3) *Planner trials:* measure time, overrides, realized service/inventory and adoption (Cannas et al., 2024; Trifone et al., 2026; Venkatesh et al., 2024) prospectively; published guarantees elsewhere do not transfer to InventOR.

8. Conclusion

InventOR provides auditable screening evidence for training-free replenishment proposals on 365 industrial plant-material pairs. Original policies cost about 11% less than ERP-derived parameters at slightly lower mean service, but classical baselines remain cheaper,

Plant C reverses the pooled ERP ranking and shortage pricing changes the conclusion. Six-run replay preserves aggregate rankings on one 338-pair panel, not independent datasets. Static checks and capacity clipping miss joint trajectory constraints; S2 supplies separate, conditional admission-rule support rather than a service guarantee. Comparator-aware cost/service screening and explicit order-admission checks are the practical result. Source-data validation, prospective planner studies and external validation remain necessary.

Appendix A. Formal Definitions

Reference calculation.. These are standard textbook calculations (Axsäter, 2015; Silver et al., 2017). For pair i, $\bar{d}_i, \sigma_{D,i}$ describe pre-cutoff working-day demand; $L_i, \sigma_{L,i}$ describe recorded lead times; K_i is ordering cost, h_i annual holding cost per unit, and z_i the normal service-target quantile. Set $Q_i = \max\{\text{MOQ}_i, \lceil\sqrt{2K_i \cdot 250\bar{d}_i/h_i}\,\rceil\}$, $\sigma_{LTD,i} = \sqrt{L_i^{\text{eff}}\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, $SS_i = \lceil z_i\sigma_{LTD,i}\rceil$ and $R_i = \lceil\bar{d}_i L_i^{\text{eff}} + SS_i\rceil$. Here $L_i^{\text{eff}} = L_i$ for History-only and $L_i + \max(\text{ERP SLT}_i, 0)$ for SLT-informed and ERP-floor; the latter also floors SS_i at ERP safety stock. SAP-derived uses ERP safety stock, the same Q_i, and ERP safety lead time in its reorder point. Projection enforces $SS_i + Q_i \le \texttt{max_storage_units}_i$ when MOQ plus safety stock fits.

Simulation.. On each post-cutoff working day, due receipts arrive first, demand is served, unmet demand is lost, and at most one order is placed when inventory position (on hand plus on order) is at or below R_i. It arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. Cost is $TC_i = h_i\bar{I}_i T/250 + K_i N_i$, where $\bar{I}_i$ is mean on-hand stock over T working days and N_i counts orders. Fill is served divided by demanded units. Hard-limit orders are $\min\{Q_i, \text{capacity}_i - \lfloor\text{position}_i\rfloor\}$ if positive, even when below MOQ; no order is placed otherwise. Observed

opening stock is not clipped.

Author Contributions

Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review and editing: Aris Dressino.

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Declaration of Competing Interest

The author reports an employment relationship with an industrial manufacturer (name withheld). The author declares no other competing interests.

Data and Code Availability

Raw operational data are confidential. A masked panel contains the same 233,078 records, with pseudonymous identifiers and secret per-pair quantity and price factors. The panel, rescaled frozen policies, three masked generations, sanitized provenance, scoring code, aggregate evidence and manuscript sources are at `github.com/DressPD/inventor_public`. The archive version DOI is `https://doi.org/10.5281/zenodo.23627916` (software v1.3.0). That archive preserves the evaluated evidence; subsequent manuscript reporting corrections do not amend its frozen source snapshot. Supplementary File S1 contains the prompt.

Runner-hash amendments are recorded separately from original call provenance, with retained original hashes and amendment histories; they must not be read as original execution hashes. Model identity (Claude Sonnet 4.6) is operator-declared; runtime identifier and sampling settings are unavailable.

Supplementary File S2 supplies anonymous methods, code, masked inputs, frozen policies, tests and separate strict-admission results. This later working support is distinct from the unchanged archives; it supports replay, not proprietary inference reconstruction.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

OpenCode AI-assisted tools, including Claude Sonnet 5.5 and GPT-6.1 Sol, assisted language editing, restructuring, literature search and reference selection, consistency and citation checks, LaTeX preparation and analysis-script development. These preparation tools are distinct from the operator-declared Claude Sonnet 4.6 deployment evaluated in the study. The author retains responsibility for reviewing and verifying the resulting text and code and for the content of the final article. Figures are deterministic diagrams and data plots rendered from source files, not image-model outputs.

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